



MD Redwan Hossain

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🏠 Google Scholar

👤 PROFILE

A passionate and ambitious individual with expertise in data analytics, machine learning, AI, and web development. I am committed to mastering Python, SQL, statistical modeling, ML & AI. Skilled in data preprocessing, feature engineering, and model optimization using Scikit-Learn, PyTorch, and Power BI. Brings a strong mix of technical expertise in Data Science and web development background to deliver practical, data-driven business solutions.

🏢 PROFESSIONAL EXPERIENCE

Master Thesis Student (Title: Optimization of Integrated Timetabling and Vehicle Scheduling for Mixed Electric and Conventional Bus Fleets.) 2025/11 – 2026/05 | Erlangen

Department Mathematik

- **Master's thesis:** Developed an optimization-based approach for efficient and sustainable public transport planning with mixed fleets of electric and conventional buses.
- **Core challenge:** Addressed the operational limits of electric buses, including battery range, charging time, and charging infrastructure at terminals or depots.
- **Research approach:** Compared three methods: a greedy heuristic, a hybrid MIP model, and a fully integrated charging-aware mixed-fleet MIP.
- **Key result:** The integrated model performed best by jointly considering routing, battery feasibility, charging decisions, and capacity constraints.
- **Technical tools:** Applied mathematical optimization and data science using **Python, Pyomo, Gurobi, and py5**.

Web Developer 2020/03 – 2021/12 | Dhaka, Bangladesh

Aarashi Online Store

- **API Development:** Developed scalable Web APIs and tested them using Postman.
- **Database Design:** Designed and developed the database structure.
- **UI Development:** Created user interfaces integrating RESTful APIs and HttpClient.
- **Documentation:** Produced comprehensive documentation for the application.

Skills: JavaScript Framework, Responsive Design, Backend Development, API Development, Postman, SQL Server, Data Warehouse, DevOps, Cloud Service, Modern Frameworks

Software Engineer Trainee 2019/12 – 2020/02 | Dhaka, Bangladesh

Unlocklive IT

- **API Development:** Assisted in developing and testing scalable Web APIs.
- **Database Design:** Participated in designing and implementing databases.
- **UI Development:** Contributed to creating user interfaces using RESTful APIs.

Skills: JS, CSS, Laravel, PHP, MySQL

🎓 EDUCATION

M.Sc. in Data Science (Major Machine Learning & AI) 2022/04 – 2026/05 | Erlangen, Germany

University of Erlangen-Nuremberg

Pattern Recognition | Mathematics of Learning | Pattern Analysis | Machine Learning in Signal Processing | ML for Time Series | Deep Learning | Advanced Anomaly Detection for ML | High-dimensional Statistics | Control Machine Learning & Numerics | Statistics | Business Intelligence

PROJECTS

Using Machine Learning to Classify Students and Predict Student Purchases

- Developed a machine learning pipeline to predict student conversion from free to paid plans using engagement data.
- Trained and evaluated Logistic Regression, KNN, SVM, Decision Tree, Random Forest
- Applied feature engineering (VIF-based feature selection, categorical encoding, scaling) to improve model performance.
- Achieved strong predictive accuracy and ROC-AUC > 0.8, enabling targeted marketing and improved customer retention strategies

Heart-Disease-Prediction-Using-Machine-Learning

Developed a machine learning classification model to predict the presence of heart disease based on patient health metrics.

- Trained and evaluated multiple algorithms (Logistic Regression, KNN, SVM, Decision Tree, Random Forest) using accuracy, precision, recall, and ROC-AUC.
- Technologies used: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook.

Bulldozer-Price-Prediction-Using-Machine-Learning

- Built an end-to-end machine learning regression model to predict bulldozer auction prices using historical sales data.
- Trained and optimized models (Random Forest Regressor, Gradient Boosting) with cross-validation and hyperparameter tuning.
- Technologies used: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook.

PUBLICATIONS

SafeBirth AI: Tuned Explainable Ensembles for Maternal and Fetal Health Classification.

2025/12/22

IEEE WIECON-ECE, Accepted for Publication (2025) [Not yet live]

- **Model Development:** Designed a voting ensemble of **XGBoost, Random Forest, and Decision Trees**, achieving **98.39% accuracy**.
- **Data Engineering:** Resolved class imbalances using a hybrid **SMOTE+ROS** approach and optimized parameters via **Bayesian/GridSearch**.
- **Explainable AI:** Integrated **SHAP and LIME** to provide global and local interpretability for trustworthy clinical decision-making.
- **Performance Validation:** Conducted comparative analysis across CTGs and Maternal datasets using **stratified cross-validation** and **ROC-AUC metrics**.

Protecting Regular and Social Network Users in a Wireless Network by Detecting Rogue Access Point: Limitations and Countermeasures

2021

Taylor & Francis Group

Authored a chapter in "Securing Social Networks in Cyberspace" (CRC Press, 2021), focusing on detecting and mitigating rogue access points (RAPs) in wireless networks, including experimental results and security measures.

SKILLS

Python, C++, SQL, ReactJs, Laravel

Microsoft Office 365

Pandas, NumPy, Matplotlib, Scikit-Learn, Pytorch,
Seaborn, Pytorch, PowerBI, Data Visualization, Cloud
Service, Cloud Platform, Model Deployment, Big data ,
Data Engineering, MLOps, Business Intelligence, Data
Warehouse, Statistical Analysis

LANGUAGES

English (Fluent)

IELTS 6.5

German (A2)

Bengali (Native)